FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Main)-2021

TEST DATE: 11-04-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. $W_{1-2} + W_{2-3} + W_{3-1} = -300$ ($\Delta U = 0$ in a cyclic process).

$W_{3-1} = 0$ $W_{1-2} = P(V_2 - V_1) = nR(T_2 - T_1) = 600R = 4980$ J

Therefore $W_{2-3} = -5280$ J

2. D

Sol. As the inclined plane is smooth, the sphere can never roll rather it will just slip down.
Hence, the angular momentum remains conserved about any point on a line parallel to the inclined plane and passing through the centre of the ball.

3. A

Sol. From equation

Av = constant, $v \propto \frac{1}{\text{Area of cross section}}$

$P + \frac{1}{2}\rho v^2 = \text{constant}$

If velocity increases then pressure decreases

4. C

Sol. $\frac{1}{f_\ell} = (1.5 - 1)\left[\frac{1}{30} - \frac{1}{-30}\right]$

$\Rightarrow f_\ell = 30$ cm

$P = P_1 + P_2 + P_3 = \frac{1}{30} + \frac{1}{30} + \frac{1}{15} = \frac{4}{30} = -\frac{1}{F_m}$

$\therefore F_m = -7.5$ cm

$\frac{1}{v} + \frac{1}{-60} = -\frac{2}{15} \Rightarrow \frac{1}{v} = \frac{1}{60} - \frac{2}{15} \Rightarrow v = -\frac{60}{7}$ cm

5. B

Sol. $3Tv_0 + T(v_0 - v) = 0$

$\Rightarrow 4v_0 = v$

$v_0 = \frac{v}{4}$

6. C

Sol. $P_{av} = V_{rms} I_{rms} \cos\phi$

Since, $\phi = \frac{\pi}{2}$

$\Rightarrow P_{av} = 0$.

7. B

Sol. For 1 ring,

$I_{CM} = mr^2 - m(2r/\pi)^2$;

$I = I_{CM} + m(r^2 + (2r/\pi)^2) = 2mr^2$

COM of system will be point of intersection of the two half rings.

Hence, the moment of inertia of the system is $I = 2mr^2 + 2mr^2 = 4mr^2$

8. D

Sol. $v = 2t - 3$; $v = 0$ at $t = 1.5$ s; $x(t = 1.5) = -0.25$ m; $x(t = 3) = 2$m ; $x(t = 0) = 2$m

Therefore distance travelled in 3 sec = 2(2 + 0.25) = 4.5 m

9. A

Sol. Work function, ϕ = 12400/5000= 2.48 eV

hc/λ – Work function = KE max = 3eV

hc/λ = 5.48 eV $\Rightarrow \lambda$ = 2260 Å

10. A

Sol. $K_A A\frac{dT_A}{dx} = K_B A\frac{dT_B}{dx}$

$200 \times \tan 60° = K_B \tan 45°$

$K_B = 200\sqrt{3} = 200 \times 1.732 = 346.40$ J/m-sec-°C

11. B

Sol. Potential difference, $V_A - V_P = \int (\vec{v} \times \vec{B}).d\vec{l} = 2B_0Rv$

12. D

Sol. Using $v = \sqrt{\frac{T}{\mu}}$, μ = mass per unit length of the rope

If v_t is velocity at top and v_b is velocity at bottom then

$$\frac{\lambda_t}{\lambda_b}=\frac{v_t}{v_b}=\sqrt{\frac{T_t}{T_b}}=\sqrt{\frac{(M+m)g}{mg}}=\sqrt{\frac{M+m}{m}}$$

$$\lambda_t=\lambda\sqrt{\frac{M+m}{m}}$$

13. B

Sol. Time of flight = $\frac{2u\sin\theta}{g}=\frac{2\times 20\times\frac{1}{2}}{10}=2\text{sec}$

After 1 sec, the projectile is at maximum height.

Maximum height $(H)=\frac{u^2\sin^2\theta}{2g}=\frac{20^2\times\left(\frac{1}{2}\right)^2}{2\times 10}=5\text{ m}$

Range $(R)=\frac{u^2\sin 2\theta}{g}=\frac{20^2\times\frac{\sqrt{3}}{2}}{10}=20\sqrt{3}\text{ m}$

$$\overline{AB}=\sqrt{H^2+\left(\frac{R}{2}\right)^2}=\sqrt{5^2+(10\sqrt{3})^2}=\sqrt{325}=5\sqrt{13}\text{ m}$$

14. A

Sol. The cylinder begins to topple when the vertical line through CG passes beyond the base of cylinder.

$$\tan\theta=\frac{r}{\left(\frac{h}{2}\right)}=1$$

or $r=\frac{h}{2}=10\text{cm}$

15. B

Sol. $\beta_2/\beta_1=\lambda_2/\lambda_1=2$

Fringe width of λ_2 = 2* Fringe width of λ_1

$2n^{th}$ order maxima of λ_1 coincide with n^{th} order maxima of λ_2

16. C

Sol. $G=\frac{0.15}{0.05}=3\Omega$

$$S=\frac{i_gG}{(I-i_g)}=\frac{0.05\times 3}{(100-0.05)}=0.0015\ \Omega$$

17. B

Sol. For permanent magnet we prefer a material with high retentivity (so as to make a stronger magnet) and high coercivity (so that magnetization can not be wiped out easily). For electromagnet we prefer high saturated magnetism low coercivity and least possible area of hysteresis loop so that electromagnet develops high magnetization, is easily demagnetized and energy loss in a magnetization cycle is least. Therefore, P is suitable for making permanent magnet and Q for making electromagnet.

18. D

Sol. $y=\overline{A+B}\cdot B$

19. C

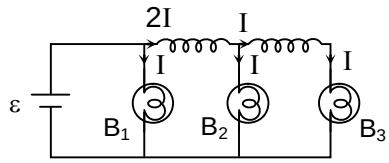
Sol. Momentum and charges are equal and opposite, so sense of the motions are opposite.

$\therefore R = \dfrac{mv}{qB}$

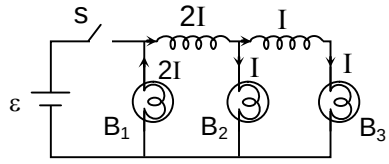
$\Rightarrow$ Their radii are equal but angular velocities will be different.

20. B

Sol. Currents before opening the switch 'S'



Currents just after opening the switch 'S'



SECTION – B

21. 00000.16

Sol. Given $y(x,t) = \dfrac{0.8}{[(8x+10t)^2+5]}$

y will be maximum when denominator is minimum.

So, $\left[(8x+10t)^2+5\right]$ must be minimum. It will be minimum when x = 0 and t = 0

So, $y_{max} = \dfrac{0.8}{0+5}$

or, y_{max} = 0.16 m

22. 00004.88

Sol. When circular scale lies below reference line error is called positive and when lies above reference line error is called negative

error = least count × no. of division which differs from reference line

or error = + 0.01×12 = 0.12 mm

Actual diameter = Reading – error

or Actual diameter = 5 mm – [+ 0.12 mm]

or Actual diameter = 4.88 mm

23. 00002.50

Sol. Given $f_1 - f_2 = 5$

or $\left(\dfrac{v}{v-v_s}\right)f - \left(\dfrac{v}{v+v_s}\right)f = 5$ or $\left[\dfrac{1}{(1-v_s/v)} - \dfrac{1}{(1+v_s/v)}\right]f = 5$

or $\left[\left(1-\dfrac{v_s}{v}\right)^{-1} - \left(1+\dfrac{v_s}{v}\right)^{-1}\right]f = 5$ or $\left[\left(1+\dfrac{v_s}{v}\right) - \left(1-\dfrac{v_s}{v}\right)\right]f = 5$

or $\dfrac{2v_s f}{v} = 5$ or Speed of tuning fork $v_s = \dfrac{5v}{2f}$

Substituting the values, we get $v_s = \dfrac{5\times 332}{2\times 332} = 2.5 \text{ m/s}$

24. 00000.28

Sol. $I = (I_o/2)\cos^2 30^o \times \cos^2 30^o$

$= \frac{I_0}{2} \times \frac{3}{4} \times \frac{3}{4} = \frac{9I_0}{32}$

25. 00072.90

Sol. $v = \sqrt{\frac{B}{\rho}}$

$\Rightarrow B = \rho v^2 = 2.5 \times 10^3 \times (5.4 \times 10^3)^2 = 72.90 \times 10^9$ N/m^2 = 72.90 (GPa)

26. 00033.33

Sol. From Newton's law of cooling

$$\left(\frac{\theta_1 - \theta_2}{t}\right) = k\left[\frac{\theta_1 + \theta_2}{2} - \theta_0\right]$$

In the first case, $\left(\frac{50-45}{5}\right) = k\left[\frac{50+45}{2} - \theta_0\right]$ …(i)

In the second case, $\left(\frac{45-41.5}{5}\right) = k\left[\frac{45+41.5}{2} - \theta_0\right]$ …(ii)

From (i) and (ii)

$\theta_0 = 33.33°C$

27. 00040.00

Sol. COP = $T_2/(T_1-T_2)$=1.2 ; T_1/T_2 = 11/6 ; Q_1 = 110 kJ/ min ; W = 50 kJ/ min

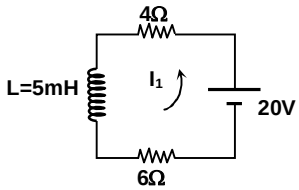
W = (5/6) kJ/ sec; $\Rightarrow$ 5/6 units per hour

Total cost = (5/6) $\times$ 6 $\times$ 8 = 40 Rs per day

28. 00002.21

Sol. $I_1 = \frac{20}{10}\left(1 - e^{-\frac{t}{5\times10^{-4}}}\right)$

$= \frac{3}{2} = 1.5$ A

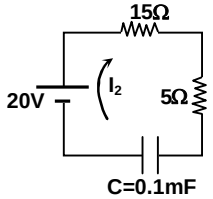


$I_2 = \frac{20}{20} e^{-\frac{t}{2\times10^{-3}}}$ = 1/√2 A= 0.71A

From superposition

$I = I_1 + I_2 = 1.50 + 0.71$

= 2.21 A



29. 00005.25

Sol. $\frac{E_1 + E_2}{E_1 - E_2} = \frac{25}{17} \quad \Rightarrow \quad \frac{E_1}{E_2} = \frac{21}{4} = 5.25$.

30. 00000.04

Sol. For dispersion without average deviation

$(n_y - 1)A = (n'_y - 1)A'$ …(i)

For no dispersion

$A(n_v - n_R) = (n'_v - n'_R)A'$ …(ii)

$\Rightarrow (n_y - 1)A\omega = (n'_y - 1)A'\omega'$

$\Rightarrow \omega = \omega' \quad [\because (n_y - 1)A = (n'_y - 1)A']$

$\Rightarrow \omega' = \omega = \frac{n_v - n_R}{n_y - 1} = \frac{1.51 - 1.49}{1.5 - 1} = \frac{0.02}{0.5} = .04$

Chemistry

PART – B

SECTION – A

31. B

Sol. Time (t) = March + April + May + June + July + August + September + October
= 31 + 30 + 31 + 30 + 31 + 31 + 30 + 31
= 245 days

$$\lambda = \frac{2.303}{245}\log_{10}\left(\frac{4}{0.5}\right) = \frac{2.303}{245}\log_{10}\left(2^3\right)$$

$$\lambda = 3\times\frac{2.303}{245}\log_{10}2 = \frac{3\times0.693}{245}$$

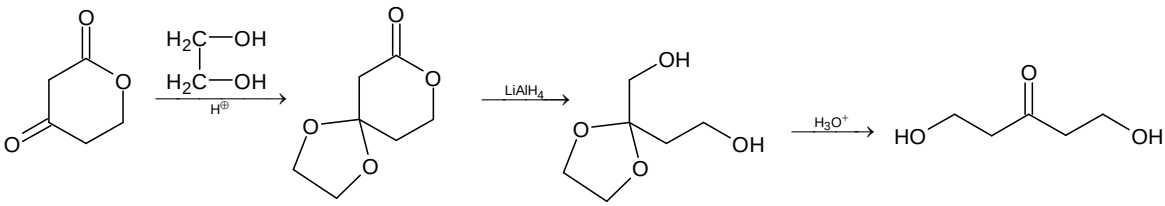
$$t_{1/2} = \frac{0.693}{\lambda} = \frac{0.693}{\left(3\times\frac{0.693}{245}\right)} = \frac{245}{3} = 81.67\,\text{days}$$

32. D

Sol. Fact

33. A

Sol.



34. B

Sol. Solubility (S) $= \left[X^{+3}\right] + \left[X(OH)_4^-\right]$

$$S = \frac{K_{sp}}{\left[OH^-\right]^3} + K_c\left[OH^-\right]$$

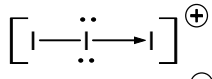
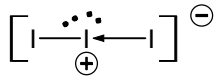
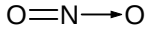
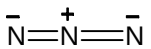
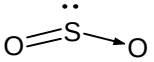
Differentiate above with respect to $\left[OH^-\right]$, for minimum solubility

$$\frac{dS}{d\left[OH^-\right]} = \frac{-3K_{sp}}{\left[OH^-\right]^4} + K_c = 0$$

$$\left[OH^-\right] = \left(\frac{3K_{sp}}{K_c}\right)^{1/4}$$

35. D

Sol.

I_3^+		v-shape
I_3^-		Linear
NO_2^+		Linear
N_3^-		Linear
SO_2		v-shape

36. C

Sol. (A) Copper pyrite $\rightarrow CuFeS_2$

(B) Galena $\rightarrow PbS$

(C) Limonite $\rightarrow Fe_2O_3.3H_2O$

(D) Copper glance $\rightarrow Cu_2S$

37. B

Sol.

	$3A$ +	$5B$	$\longrightarrow$	$4C$	$\Delta H = -600$ kJ
$t = 0$	9	20		0	
t	$9-9=0$	$20-\frac{5}{3}\times 9$			

= 5 mole

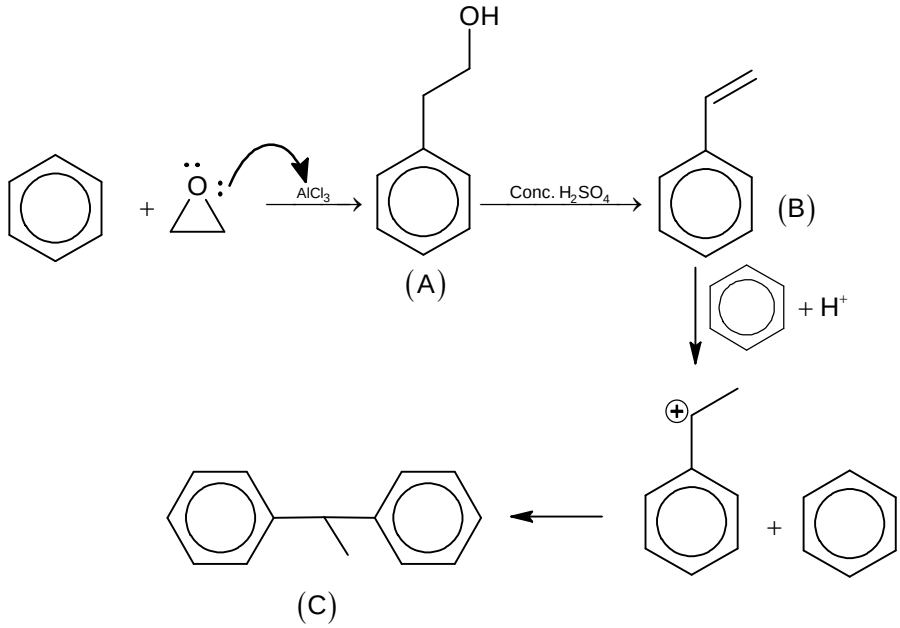
A will be the limiting reagent.

Final change in enthalpy $= -\frac{600}{3}\times 9$

= - 1800 kJ

38. C

Sol.



39. D

Sol. In BCC number of atoms of $A = 1+\frac{1}{8}\times 8 = 2$

B is inside A so volume unoccupied by A in unit cell $= 2\left(\frac{4}{3}\pi r_A^3 - \frac{4}{3}\pi r_B^3\right)$

$4r_A = a\sqrt{3}$ (By diagonal)

$$\text{Ratio} = \frac{2\times\frac{4}{3}\pi\left(r_A^3 - r_B^3\right)}{a^3} = \frac{2\times\frac{4}{3}\pi\left(r_A^3 - \left(\frac{1}{4}r_A\right)^3\right)}{\left(\frac{4r_A}{\sqrt{3}}\right)^3}$$

$$= \left(\frac{\frac{8\pi}{3} \times \frac{63}{64} r_A^3}{64 r_A^3} \times 3\sqrt{3} \right) = \frac{63\sqrt{3}}{512}$$

40. C

Sol. In 4th period s, p, d will present.
So number of orbitals 1 + 3 + 5 = 9
In each orbital maximum number of electron = 5
Hence, total number of elements in 4th period = 5 × 9 = 45 elements.

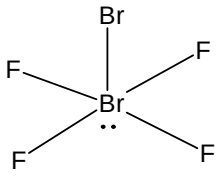
41. B

Sol. In i, iii and v reaction NH_4NO_3 is also formed with metal nitrate.

42. C

Sol.

BrF_5



sp^3d^2

square pyramidal

43. D

Sol. $\underset{(X)}{SiO_2} + 2C \longrightarrow Si + \underset{\text{(Poisonous gas stable)}}{2CO}$

44. D

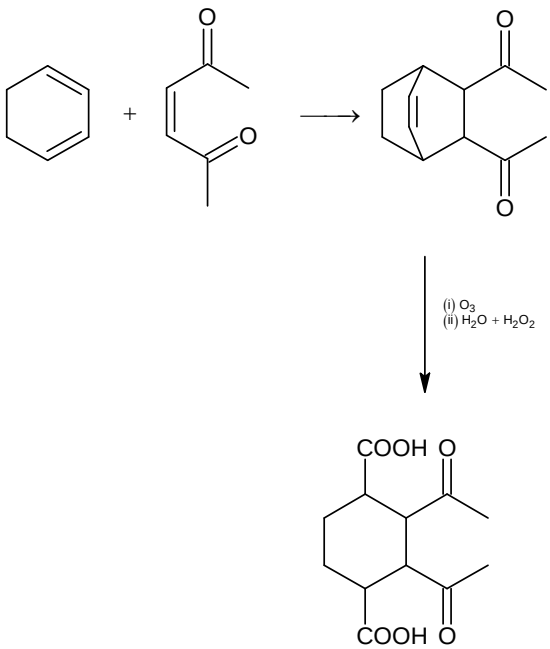
Sol. Electronegativities are H – 2.1, P – 2.1, N – 3 , As – 2.0, Sb – 1.9
Polarity $\propto$ difference in electronegativity order is

$NH_3 > SbH_3 > AsH_3 > PH_3$

Difference in electronegativity	0.9	0.2	0.1	0

45. A

Sol.

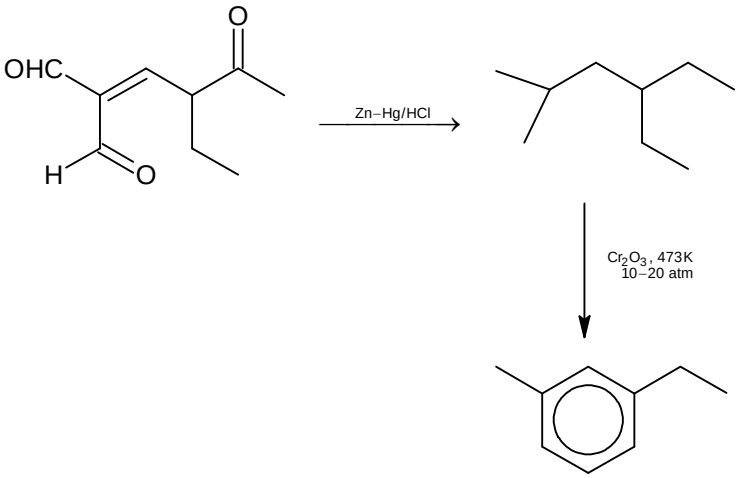


46. A

Sol. Benzyl and alkyl position is reactive toward nucleophilic substitution reaction. Phenol will not react with $SOCl_2$.

47. A

Sol.



48. D

Sol. Both are correct statement. Fact.

49. C

Sol. Carbon and NaOH does not react with each other.

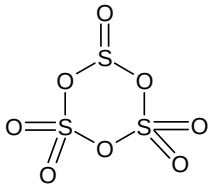
50. B

Sol. This test is used for proteins.

SECTION – B

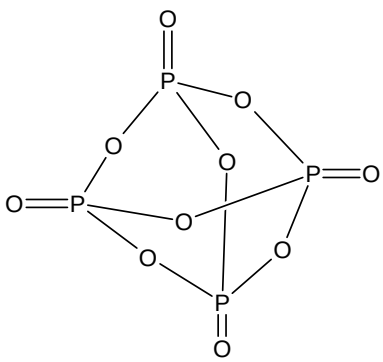
51. 00002.25

Sol. γ form of SO_3 is S_3O_9



3 S – O – S bonds

P_4O_{10}



6 P – O – P bonds
x = 3 + 6 = 9

$$\text{Ratio} = \frac{x}{4} = \frac{9}{4} = 2.25$$

52. 00001.20

Sol. $CoCl_3.xNH_3$

$$\left[CoCl_{3-y}.(NH_3)_x\right]Cl_y \rightleftharpoons \left[CoCl_{3-y}(NH_3)_x\right]^{+y}_{(aq)} + yCl^-_{(g)}$$

$$i = 1 + y$$

$$\Delta T_f = iK_f m$$

$$0.558 = (1+y)0.558 \times 0.1$$

y = 2

Hence, compound will be

$$\left[Co(NH_3)_5 Cl\right]Cl_2$$

x = 5

$$\text{Ratio} = \frac{\text{C.N. of cobalt}}{x} = \frac{6}{5} = 1.20$$

53. 00004.95

Sol. $2\text{mg of } CaCl_2 \equiv \frac{100}{111} \times 2 \text{ mg of } CaCO_3$

$= 1.8 \text{ mg of } CaCO_3$

$2\text{mg of } MgCl_2 \equiv \frac{100}{95} \times 3 \text{ mg of } CaCO_3$

$= 3.15 \text{ mg of } CaCO_3$

One litre of hard water contains 1.8 + 3.15
= 4.95

54. 00067.20

Sol. Volume strength $= 5.6 \times$ Normality

$= 5.6 \times 12$
= 67.20

55. 00001.25

Sol. Polar molecules will show deflection near charged comb.

$a = H_2O_2, HF, NH_3, CHCl_3, C_6H_5Cl \quad (5)$

$b = O_2, CCl_4, C_6H_6, N_2 \quad (4)$

$$\frac{b}{a} = \frac{5}{4} = 1.25$$

56. 00001.76

Sol. $2CH_3COONa \xrightarrow{\text{Electrolysis}} 2NaOH + H_2 + C_2H_6 + 2CO_2$

$$CH_3COONa \xrightarrow{\text{Electrolysis}} NaOH + \frac{1}{2}H_2 + \frac{1}{2}C_2H_6 + CO_2$$

For above 1 F is required to form 2 mole gases.

Charge supplied $= i \times t = 0.965 \times 60 \times 60$ C

So moles of gases formed $= 2 \times$ Charge supplied/1F

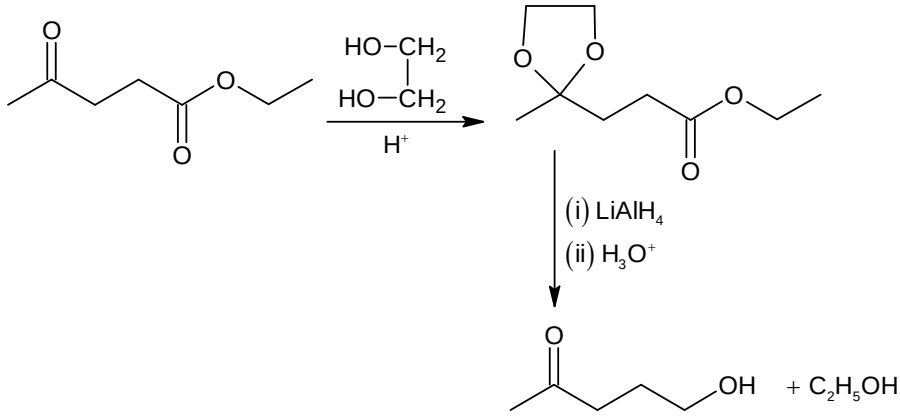
$$= \frac{2 \times 0.965 \times 60 \times 60}{1F} = 0.072$$

$$PV = nRT \Rightarrow V = \frac{nRT}{P}$$

$$V = \frac{0.072 \times 0.0821 \times 298}{1} = 1.761 \text{ lit}$$

57. 00001.50

Sol.



$$x = 3\left(LiAlH_4, \text{ Ethylene glycol, } H_3O^+\right)$$

$$y = 2(\text{Acetone, DMF})$$

$$\frac{x}{y} = \frac{3}{2} = 1.50$$

58. 00000.50

Sol. Number of correct statements = 2 (c, e)

Number of incorrect statements = 4 (a, b, d, f)

$$\frac{a}{b} = \frac{2}{4} = 0.50$$

59. 00001.60

Sol. Let A_o and B_o are initial activities of A and B respectively. After 200 min A will $1/4^{th}$ of A_o.

$t = 0 \quad A_o + B_o = 13000\ \beta - \text{particle / min} \quad \ldots$ (i)

$t = 200 \quad \frac{A_o}{4} = 12500\ \beta - \text{particle / min} \quad \ldots$ (ii)

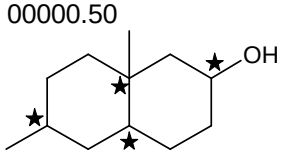
$A_o = 5000 \quad B_o = 13000 - 5000$

$= 8000$

$$\frac{B_o}{A_o} = \frac{8000}{5000} = 1.60$$

60. 00000.50

Sol.



Number of chiral carbon atoms = 4 (* marked)

Number of achiral carbon atoms = 8 (Rest are achiral C atoms)

$$\text{Ratio} = \frac{4}{8} = 0.50$$

Mathematics

PART – C

SECTION – A

61. B

Sol. From both equation $B^2 = A^{-1}$

62. C

Sol. G.M. of sub-tangent and subnormal = $\sqrt{\frac{y}{\frac{dy}{dx}} \cdot y \cdot \frac{dy}{dx}} = |y|$, ratio of length of normal to the tangent =

$$\left| \frac{y\sqrt{1+\left(\frac{dy}{dx}\right)^2}}{y\sqrt{1+\frac{1}{\left(\frac{dy}{dx}\right)^2}}} \right| = \left|\frac{dy}{dx}\right|$$

$\Rightarrow |y| = \left|\frac{dy}{dx}\right| \Rightarrow y' = \pm y \Rightarrow y = 2e^x$ or $y = 2e^{-x}$

63. B

Sol. $m_a = c \sin B$ and $d_a = 2R \cos A$

$m_b = a \sin C$ and $d_b = 2R \cos B$

$m_c = a \sin B$ and $d_c = 2R \cos C$

$\Rightarrow m_a d_a + m_b d_b + m_c d_c$

$\Rightarrow 2R \cos A \, c \sin B + 2R \cos B \, a \sin C + 2R \cos C \, a \sin B$

$\Rightarrow bc \cos A + ac \cos B + ab \cos C$

$$\Rightarrow bc\left(\frac{b^2+c^2-a^2}{2bc}\right) + ac\left(\frac{a^2+c^2-b^2}{2ac}\right) + ab\left(\frac{a^2+b^2-c^2}{2ab}\right) \Rightarrow \frac{a^2+b^2+c^2}{2}$$

64. C

Sol. $x^2 + 2x + a = 0 \Rightarrow a = -x^2 - 2x$ and $ax^2 + x + 2 = 0 \Rightarrow (-x^2 - 2x)x^2 + (x + 2) = 0$

$\Rightarrow (x + 2)(-x^3 + 1) = 0 \Rightarrow x = -2$ or 1 or $x^2 + x + 1 = 0$ when $x = -2 \Rightarrow a = 0$

$x = 1 \Rightarrow a = -3$

$\therefore$ Product of all possible values of a is 0

65. C

Sol. $(1-x^2)\frac{dx}{dy} = x(1-y) \Rightarrow \left(\frac{1}{x} - x\right)\frac{dx}{dy} = (1-y)$

$$\Rightarrow \left(\frac{1}{x} - x\right)dx = (1-y)dy \Rightarrow \ln|x| - \frac{x^2}{2} = -\frac{(1-y)^2}{2} + c$$

$\Rightarrow \ln x^2 = x^2 - (y - 1)^2 + c$

$\because$ $y(1) = 1 \Rightarrow c = -1 \Rightarrow g(x) = (y - 1)^2 = x^2 - \ln x^2 - 1$

$$\Rightarrow \lim_{x\to 1} \frac{g(x)}{(x-1)^2} = \lim_{x\to 1} \frac{x^2 - \ln x^2 - 1}{(x-1)^2} = 2$$

66. C

Sol. $(a^2 - b^2)(a - b) = 8R^3(\sin^2 A - \sin^2 B)(\sin A - \sin B)$

$\Rightarrow 8R^3(\sin^2 (3B) - \sin^2 B)(\sin 3B - \sin B) \Rightarrow 8R^3(\sin (3B) - \sin B)^2(\sin (3B) + \sin B)$

$\Rightarrow 8R^3 \sin^2 4B \sin B \Rightarrow 8R^3 \sin^2(\pi - c) \sin B \Rightarrow 8R^3 \sin^2 C \sin B \Rightarrow bc^2$

67. C

Sol. $L = \lim\limits_{x\to 0} \dfrac{3\lambda x + (\lambda-2)\left(x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!}.....\right)}{\dfrac{(\sin^{-1}(x))^3}{x^3}\cdot x^3} = \lim\limits_{x\to 0} \dfrac{x(4\lambda-2) - \left(\dfrac{\lambda-2}{6}\right)x^3.....}{x^3}$

$\because$ limit is finite $\lambda = \dfrac{1}{2}$ and $L = \dfrac{1}{4}$

68. D

Sol. $\int \sin\left(\cot^{-1}\left(\sqrt{x^2-1}\right)\right)dx = \int \dfrac{dx}{x} = \ln x + k$

$\because$ $f(x) = \ln x \Rightarrow f(e^2) = 2$

69. C

Sol. $f(x) = \int\limits_0^x e^{x-y} f'(y)\,dy - (x^2 - x + 1)e^x$ ..... (1)

$\Rightarrow f'(x) = e^x \int\limits_0^x e^{-y} f'(y)\,dy + e^x\left(1\cdot e^{-x}\cdot f'(x)\right) - (2x-1)e^x - (x^2-x+1)e^x$

$\Rightarrow \int\limits_0^x e^{x-y} f'(y)\,dy = (x^2+x)e^x$ ..... (2)

From equation (1) and (2), we get $f(x) = (2x-1)e^x$

70. C

Sol. At $x = -\dfrac{5\pi}{8}$; $y = \cos^{-1}(\cos(-2x+2)) = 2\pi - (-2x+2)) = 2x - 2 + 2\pi$

$\dfrac{dy}{dx} = 2$

71. A

Sol. $\dfrac{\sin^4\theta + \cos^4\theta}{\sin^2\theta\cos^2\theta} = 4\,\text{cosec}^2 2\theta - 2 = 2 + 4\cot^2 2\theta$

$\Rightarrow \sum\limits_{\theta=1^\circ}^{89^\circ} (2 + 4\cot^2 2\theta) = 2(89 + 4a)$

72. C

Sol. $P(A \cup B') = 0.60$; $P(A \cap B) = 0.10$

$\Rightarrow P\left(\dfrac{B}{A\cup B'}\right) = \dfrac{P(B\cap(A\cup B'))}{P(A\cup B')} = \dfrac{P(A\cap B)}{P(A\cup B')} = \dfrac{1}{6}$

73. B

Sol. $N = 12 \times 11 \times 10 = 1320$

$P = \dfrac{{}^3C_1 \cdot {}^7C_1 \cdot {}^9C_1}{1320}$

74. B

Sol. Simplified expansion will be

$2\left({}^{18}C_0 (x+y)^{18} z^0 + {}^{18}C_2 (x+y)^{16} z^2 + {}^{18}C_4 (x+y)^{14} z^4 ..... {}^{18}C_{18} z^{18}\right)$

$\Rightarrow$ Numbers of dissimilar term are = 19 + 17 + 15 ..... 1 = 100

75. D

Sol. Equation of tangent to ellipse $y = mx \pm \sqrt{2m^2 + \frac{3}{2}}$ by comparing both equations

$y = px + q$ and $y = mx \pm \sqrt{2m^2 + \frac{3}{2}}$

$\Rightarrow q = \sqrt{2m^2 + \frac{3}{2}}$, $p = m \Rightarrow \frac{p^2}{3/4} - \frac{q^2}{3/2} = -1$

76. B

Sol. $\bar{a} = 2(\bar{b} \times \bar{c}) + (\bar{c} \times \bar{a}) \Rightarrow \bar{a} \cdot \bar{a} = 1 = 2\left[\bar{a}\ \bar{b}\ \bar{c}\right] \Rightarrow \left[\bar{a}\ \bar{b}\ \bar{c}\right] = \frac{1}{2}$

$\Rightarrow \bar{a} \cdot \bar{b} = \left[\bar{a}\ \bar{b}\ \bar{c}\right] = \frac{1}{2}$ and $\bar{a} \cdot \bar{c} = 0$

77. C

Sol. Focii of ellipse are (–2, 0), (2, 0) so value of a = 1

Equation hyperbola $\frac{x^2}{1} + \frac{y^2}{3} = 1$; equation of tangent $y = mx \pm \sqrt{m^2 - 3}$

Equation of ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$; equation of tangent $y = mx \pm \sqrt{8m^2 + 4}$ for common tangent

$m^2 - 3 = 8m^2 + 4$, $m^2 = -1$ not possible

78. B

Sol. Tangent to the parabola $y^2 = 8x$ will be $y = mx + \frac{2}{m}$ this tangent passes through (2, 5)

$\Rightarrow 5 = 2m + \frac{2}{m} \Rightarrow 2m^2 - 5m + 2 = 0 \Rightarrow m = 2, \frac{1}{2}$

Angle between tangent = $\tan\theta = \left|\frac{2 - \frac{1}{2}}{1 + 2 \times \frac{1}{2}}\right| = \frac{3}{4}$; $\theta = \tan^{-1}\left(\frac{3}{4}\right)$

79. D

Sol. The determinant of coefficient $\begin{vmatrix} a & ma & 1 \\ b & (m+1)b & 1 \\ c & (m+2)c & 1 \end{vmatrix} = 0 = \begin{vmatrix} 1 & m & \frac{1}{a} \\ 1 & m+1 & \frac{1}{b} \\ 1 & m+2 & \frac{1}{c} \end{vmatrix} = 0$

$\Rightarrow \begin{vmatrix} 1 & m & \frac{1}{a} \\ 0 & 1 & \frac{1}{b} - \frac{1}{a} \\ 0 & 1 & \frac{1}{c} - \frac{1}{b} \end{vmatrix} = 0 \Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

$\Rightarrow$ a, b, c are in H.P.

80. C

Sol. $((p \wedge \sim p) \vee (q \vee \sim q)) \vee (p \to q)$

$(F \vee T) \vee (p \to q)$

$T \vee (p \to q) = T$ only option (C) is tautology

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. 00001.00

Sol. $f : A \to \left[-\frac{1}{2}, 0\right]$; $f^3(x) + x^3 + 1 = 3xf(x)$

$\Rightarrow f(x) + x + 1 = 0 \Rightarrow f(x) = -x - 1$

$\Rightarrow f(f(x)) = x$ and $f^{-1}(x) = -x - 1 \Rightarrow x = -x - 1$

$x = -\frac{1}{2}$ only one solution

82. 00005.00

Sol. $\sin(x - y) + \sin(x + y) = 2\cos x \Rightarrow \sin x \cos y = \cos x$

If $\cos x = 0 \Rightarrow \cos y = 0 \Rightarrow x + y = \pi$

If $\cos x \neq 0 \Rightarrow \tan x \cos y = 1 \Rightarrow \tan x = \sec y$ for minimum value $x = \frac{\pi}{4}$, $y = 0$

83. 00024.00

Sol. $x = 75600 = 2^4\, 3^3\, 5^2\, 7^1$

Let divisor is $2^{a_1} \cdot 3^{b_1} \cdot 5^{c_1} \cdot 7^{d_1}$ and $2^{a_2} \cdot 3^{b_2} \cdot 5^{c_2} \cdot 7^{d_2}$

Product of divisors = 60480 = $2^6 \cdot 3^3 \cdot 5^1 \cdot 7^1$

$\Rightarrow a_1 + a_2 = 6 \Rightarrow 3$ ways

$\Rightarrow b_1 + b_2 = 3 \Rightarrow 4$ ways

$\Rightarrow c_1 + c_2 = 1 \Rightarrow 2$ ways

$\Rightarrow d_1 + d_2 = 1 \Rightarrow 2$ ways

Total way $N = \frac{3 \times 4 \times 2 \times 2}{2} = 24$

84. 00004.00

Sol. Equation of circle satisfying the equation $\left|\frac{z}{z-1}\right| = \frac{1}{2}$ is $(x+1)\left(x - \frac{1}{3}\right) + y^2 = 0$; minimum distance of circle from point $\left(-\frac{1}{3}, 2\right)$ is $\frac{4}{3}$

85. 00008.00

Sol. Let (a, b, c) lies on plane 5x + 12y + 13z – 52 = 0, then distance of origin from plane is

$\sqrt{k} = \left|\frac{52}{\sqrt{25+144+169}}\right| = \left|\frac{52}{13\sqrt{2}}\right| \Rightarrow 2\sqrt{2}$

86. 00004.00

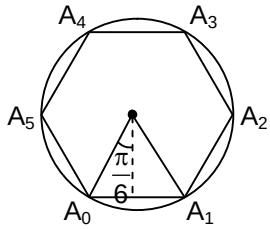
Sol. Line $\frac{x-2}{a} = \frac{y-b}{1} = \frac{z-4}{4}$ intersect the xy plane at p(2 – a, b – 1, 0)

$\because$ Point p satisfy the equation of circle

$\Rightarrow (2-a)^2 + (b-1)^2 - 8(2-a) + 2(b-1) - 1 = 0$
$\Rightarrow (a+2)^2 + b^2 = 18$
$\therefore\ a = -2 + 3\sqrt{2}\cos\theta\ ;\ b = 3\sqrt{2}\sin\theta$
$\Rightarrow a + b = -2 + 3\sqrt{2}(\cos\theta + \sin\theta)$
Maximum (a + b) = 4

87. 00006.00

Sol. $A_0A_2 = A_0A_4 = A_2A_4 = 2.1 \sin 60° = \sqrt{3}$
$\Rightarrow (p+q)r = (2\sqrt{3})\sqrt{3} = 6$



88. 00002.00

Sol. Let equation of tangent to parabola $ty = x + 4t^2$ ..... (1)

Midpoint of AB is (h, k) then AB will be $\frac{xk + yh}{2} - c^2 = hk - c^2$ ..... (2)

By comparing (1) and (2), we get
$\Rightarrow k^2 = -2h \Rightarrow y^2 = -2x$ length of latus rectum = 2

89. 00002.66
(Range 00002.66 – 00002.67)

Sol. Equation of pair of angle bisector will be $3y^2 - 8xy - 3x^2 = 0$ ..... (1)

(As slope of first bisector is 3 then slope of second is $-\frac{1}{3}$)

Equation of angle bisector of $ax^2 + by^2 + 2hxy = 0$ will be $\frac{x^2 - y^2}{a-b} = \frac{xy}{h}$ ..... (2)

By comparing both equation $\left|\frac{a-b}{h}\right| = \frac{8}{3}$

90. 00001.50

Sol. Using A.M. $\geq$ H.M. and $a^2 + b^2 + c^2 \geq ab + bc + ca$

$$\frac{1}{ab+1} + \frac{1}{bc+1} + \frac{1}{ca+1} \geq \frac{9}{3+ab+bc+ca} \geq \frac{9}{3+a^2+b^2+c^2} \geq \frac{9}{6}$$